BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Storage Units and Conversions

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What should guide the choice of a storage device?

2. When does a compiler translate source code?

A) When a key is pressed
B) Only after the program ends
C) Before the program runs
D) Only while each line runs

3. For scores = [5, 1, 3], total starts at 0 and each element is added once. Give the successive totals and the final result.

4. Use decimal units. Convert 5 TB to GB, then state how many TB are in 1 PB.

Today's objectives

Recall bit, nibble, byte and decimal storage-capacity units. Convert correctly between bits, bytes, kB, MB and GB.

Key vocabulary - Read, cover, recall

Bit
A binary digit, either 0 or 1.

Nibble
Four bits.

Byte
Eight bits.

Kilobyte
1000 bytes using decimal SI units.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Small units	A bit stores one binary digit; a nibble is four bits and a byte is eight bits.
Larger units	For GCSE storage-capacity questions here: 1 kB = 1000 bytes; 1 MB = 1000 kB; 1 GB = 1000 MB. State assumptions where relevant.
Conversion direction	Moving to a smaller unit multiplies; moving to a larger unit divides. Bits and bytes require a factor of eight.
Capacity calculations	Convert all sizes to the same unit before dividing total capacity by file size.

I Do - One worked example

Convert 4 MB into kB and then into bytes using decimal units. [4 marks]

1. State the conversion factor

2. 1 MB = 1000 kB and 1 kB = 1000 bytes.

3. 4 x 1000 = 4000 kB.

4. 4000 x 1000 = 4,000,000 bytes. [4]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Convert 3 GB into MB and kB. Show both steps. [4]

2. Convert 56 bits to bytes and explain the operation. [3]

3. A 5 GB drive stores files of 250 MB each. Calculate how many fit. [4]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Display the number of leftover bits when 43 bits are grouped into bytes.

```
bits = 43
leftover = ...
print(leftover)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Convert 2.5 MB into kB using decimal units. [2]

Q3. Calculate the number of bits in 12 bytes. [2]

Q4. A device has 6 GB free. Each file uses 750 MB. Calculate the maximum whole files it can store. [3]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An algorithm only gives reliable results when its inputs are handled as specified. ☐ T ☐ F

2. A bit stores one binary digit; a nibble is four bits and a byte is eight bits. ☐ T ☐ F

3. For GCSE storage-capacity questions here: 1 kB = 1000 bytes; 1 MB = 1000 kB; 1 GB = 1000 MB ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The application's requirements, including capacity, speed, cost and physical use. ☐

Q2 C - Before the program runs ☐

Q3 5, 6, 9; final total 9. ☐

Q4 (a) 5,000 GB; 1 PB = 1,000 TB. ☐

Code First - Model

```
bits = 43
leftover = bits MOD 8
print(leftover)
```

Output: 3

We Do - Check the model

1. 3 x 1000 = 3000 MB [2]; 3000 x 1000 = 3,000,000 kB [2]. ☐

2. There are 8 bits in a byte [1]; 56 / 8 [1] = 7 bytes [1]. ☐

3. 5 GB = 5000 MB [1]; 5000 / 250 [1] = 20 files [1], assuming no overhead [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 2.5 x 1000 [1] = 2500 kB [1]. ☐

Q3 12 x 8 [1] = 96 bits [1]. ☐

Q4 6 GB = 6000 MB [1]; 6000 / 750 [1] = 8 whole files [1], ignoring overhead. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: